

The convex hull of a compact set in a finite real coordinate space

MathlibAnnex.isCompact_convexHull_pi

theorem

Realizes the ordinary convex hull as a finite union of compact images of bounded-length convex combinations.

Statement

If I is finite and $K \subseteq \mathbb{R}^I$ is compact, then $\text{conv}(K)$ is compact. The set K may be empty.

Assumptions

The coordinate space is real and has finitely many coordinates. Only compactness of K is assumed; no convexity or nonemptiness of K is required.

Conclusion

The ordinary convex hull itself is compact. No closure is added to the statement. In particular the conclusion includes $\text{conv}(\emptyset) = \emptyset$.

The finite dimension is used to bound the number of points in a convex representation. Compactness then comes from finitely many parameter spaces, not from an unrestricted union over all possible lengths.

Proof route

Bound the length of every convex combination by the dimension plus one, then take the finite union of the corresponding compact images.

▼ Proof steps

1. Write $d = |I|$. For $0 \leq k \leq d + 1$, put

$$\begin{aligned}\Delta_k &= \{t \in \mathbb{R}^k : t_i \geq 0, \sum_{i=1}^k t_i = 1\}, \\ F_k : \Delta_k \times K^k &\longrightarrow \mathbb{R}^I, \quad F_k(t, x) = \sum_{i=1}^k t_i x_i, \\ C_k &= F_k(\Delta_k \times K^k).\end{aligned}$$

For $k > 0$, Δ_k is closed and $0 \leq t_i \leq 1$, so it is compact in finite dimensions. For $k = 0$, it is empty and hence compact. The finite product $\Delta_k \times K^k$ is compact, and F_k is continuous as a finite sum of scalar multiplications. Thus every C_k is compact.

2. Every C_k lies in $\text{conv}(K)$ by the defining property of convex combinations. Conversely, the affinely independent representation of a point in the convex hull provides finitely many points in K with positive weights summing to 1. An affinely independent family in a d -dimensional space has at most $d + 1$ points: after choosing one point, its difference vectors are linearly independent and there are at most d of them. Reindexing the representation by $1, \dots, k$ puts the point in C_k for some $k \leq d + 1$.

3. Thus

$$\text{conv}(K) = \bigcup_{k=0}^{d+1} C_k.$$

The right side is a finite union of compact sets and hence compact. For $k = 0$, the empty weight sum is 0, so Δ_0 is empty. If K is empty, K^k is empty for every $k > 0$; therefore all the images are empty, and the same union proves the empty-set case.

Main citations

- Exact declaration and its source — `MathlibAnnex.isCompact_convexHull_pi`
- Compact parameter spaces and their convex-combination maps — `MathlibAnnex.fixedConvexCarrier`
- Bounded-length convex representations — `MathlibAnnex.exists_bounded_convex_representation`
- The finite union equals the convex hull — `MathlibAnnex.convexHull_eq_iUnion_fixedConvexImage`

Lean source signature (exact)

```
theorem isCompact_convexHull_pi {ι : Type*} [Fintype ι]
  (s : Set (ι → ℝ)) (hs : IsCompact s) :
  IsCompact (convexHull ℝ s)
```

In the source	Mathematical meaning
<code>{ι : Type*} [Fintype ι]</code>	The finite coordinate set I ; its real function space is $\mathbb{R}^I$.
<code>(s : Set (ι → ℝ)) (hs : IsCompact s)</code>	The set $K \subseteq \mathbb{R}^I$ and the assumption that K is compact. No nonemptiness or convexity is required.
In the source	Mathematical meaning
<code>convexHull ℝ s</code>	The ordinary real convex hull $\text{conv}(K)$, without adding a closure.
<code>IsCompact (convexHull ℝ s)</code>	The conclusion that $\text{conv}(K)$ is compact, including the empty case.

▼ Exact surrounding binder context (separate excerpt)

```
noncomputable section

open Set Finset
```

open scoped BigOperators

namespace MathlibAnnex

▼ **Exact content identity**

Declaration: MathlibAnnex.isCompact_convexHull_pi

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